

DFT Periodicity Analysis of Stop-Codon-Free Regions (SCFRs)

Across Genomic Categories in 7 Primate T2T Genome Assemblies

A visual and comparative analysis of frequency-density and frequency–amplitude–count plots for ≥ 300 bp SCFRs, generated across seven annotation categories (CDS, UTR, introns, pseudogenes, RNA genes, intergenic, and antisense-intergenic) in human, bonobo, chimpanzee, gorilla, Bornean orangutan, Sumatran orangutan, and gibbon.

Scope of this report. This report is based entirely on the 98 rendered PDF plots supplied (7 species $\times$ 7 categories $\times$ 2 plot types), plus the peak/dip annotations already drawn onto the density plots by your pipeline. No raw per-sequence frequency/amplitude data were available for this pass — several conclusions below are therefore **read directly off the rendered curves** and are flagged as such. Section 7 lays out exactly what raw-data analysis would be needed to convert each observation into a confirmed result.

How to read what follows. Observations are things visible in the plots. Interpretation is the most defensible reading of that observation, tied to what is biologically known about the region type. Speculation is explicitly flagged as such, with the reasoning that motivates it and the check needed to test it.

1. What the plots are showing

For each category, two complementary views were generated per species:

- **Frequency density plot** (X = DFT frequency 0–0.5, Y = density): a smoothed distribution of the dominant frequency found per SCFR, pooled across all qualifying SCFRs (≥ 300 bp) in that category. Two local maxima (Peak1, Peak2) and the dip between them are auto-annotated by your script, along with the density gap between the two peaks ("Peak diff").
- **Frequency–amplitude–count heatmap** (X = frequency bin, Y = amplitude bin, color = number of SCFRs in that bin): shows not just where the dominant frequency sits, but how strong (high-amplitude) the periodic signal is, and how many sequences contribute to each combination.

A DFT frequency of **0.33** ($\approx 1/3$) corresponds to a 3-bp repeat — codon-level periodicity, the signature of triplet reading-frame constraint. A frequency of **0.2** corresponds to a 5-bp repeat period, and its second harmonic folds back to **0.4** for a real-valued sequence signal — so a 0.2/0.4 twin-peak is the fingerprint of an underlying period-5 repeat element, not a coding signal. A peak near **0** reflects a slowly-varying, near-constant compositional trend (long-range GC/AT skew) rather than any short-period repeat.

Density-plot **peak height** is a within-category, within-species probability density (area under each curve integrates to 1), so height is directly comparable *across species within one category*. Comparing absolute height *across categories* is also informative here because all panels share the same 0–0.5 x-axis and were plotted at comparable scales — a much taller, narrower peak means the SCFRs in that category converge tightly on one periodicity (low compositional diversity, strong shared constraint); a low, broad, multi-modal curve means the pooled SCFRs are a heterogeneous mixture of many different periodicities (weak or absent shared constraint).

2. Category-by-category observations

2.1 CDS-overlapping SCFRs — the positive control

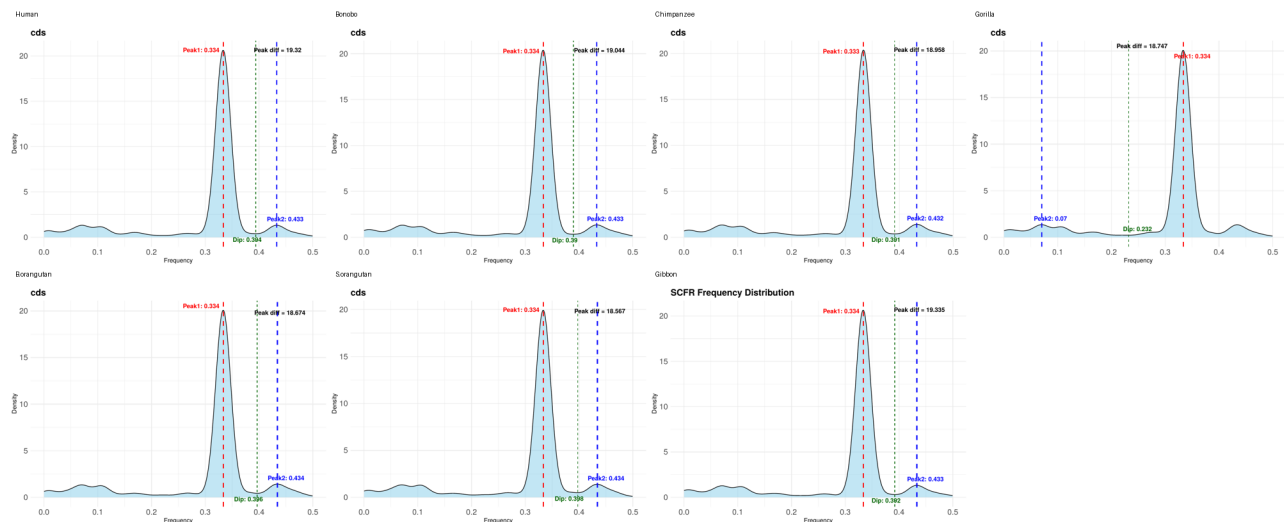


Figure 1. CDS-overlapping SCFR frequency density, all 7 species.

Observation. Every species shows a single, extremely sharp, tall peak at $f = 0.333\text{--}0.334$ with a much smaller shoulder near 0.43. Peak height ($\approx 19\text{--}21$) and peak dominance ("Peak diff" 18.6–19.3, see Table 1) are essentially identical across all 7 species — the tightest, most invariant signature in the whole dataset.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.334	0.433	0.394	19.32

Bonobo	0.334	0.433	0.390	19.04
Chimpanzee	0.333	0.432	0.391	18.96
Gorilla	0.334	0.07*	0.232*	18.75
B. orangutan	0.334	0.434	0.396	18.67
S. orangutan	0.334	0.434	0.398	18.57
Gibbon	0.334	0.432	0.392	19.34

**Gorilla's auto-annotated "Peak2" landed on a minor low-frequency bump (0.07) rather than the ~0.43 shoulder every other species shows — almost certainly a peak-picking artifact (gorilla's 0.43 shoulder is visibly present in Fig. 1 but slightly shorter than the low-frequency ripple), not a biological difference. Flagged here rather than silently corrected.*

Interpretation. This is the expected result and functions as the positive control for the whole analysis: true, mature protein-coding sequence gives an almost perfectly invariant period-3 signal regardless of species, because the triplet code and translational reading-frame constraint are universal. Any other category's approach toward this exact shape (tall, narrow, centered on 0.333) is the operational definition of "looking coding-like" used throughout this report.

The small 0.43 shoulder is a known secondary harmonic-like feature reported in prior codon-usage DFT work and most likely reflects non-uniform codon usage bias (some amino acids/codons recur with a slightly different effective period than a pure alternating triplet) rather than a second biological process. It is consistent across species, so it is a property of coding sequence composition in general, not a primate-specific feature.

2.2 Pseudogene-overlapping SCFRs

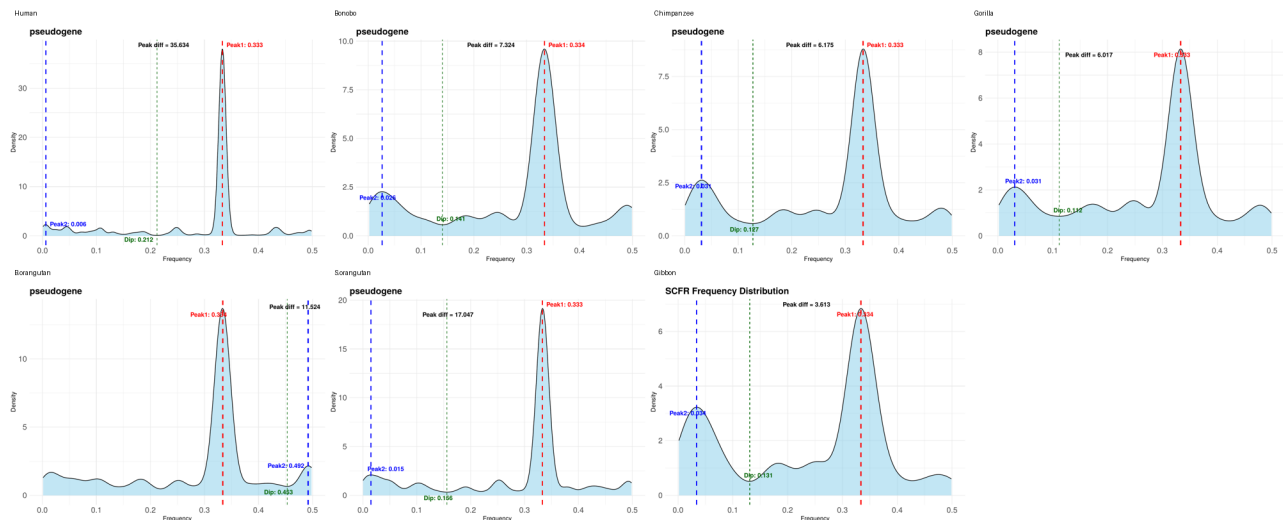


Figure 2. Pseudogene-overlapping SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak height	Peak2 (freq)	Peak diff
Human	0.333	≈35	0.006	35.63
Bonobo	0.334	≈9.5	0.026	7.32
Chimpanzee	0.333	≈8.7	0.031	6.18
Gorilla	0.333	≈8.0	0.031	6.02
B. orangutan	≈0.334	≈13	0.492	11.52
S. orangutan	0.333	≈19	0.015	17.05
Gibbon	0.334	≈7.0	0.034	3.61

Observation. All 7 species retain the CDS-like peak at 0.333–0.334 — expected, since most annotated pseudogenes are decayed copies of once-functional, still partly triplet-structured genes. But peak *height* varies roughly 5–10× across species, with human by far the tallest and sharpest (height ≈35, nearly double the next highest), and no clean phylogenetic gradient (Sumatran orangutan is 2nd-highest, ahead of chimp and gorilla, which are phylogenetically closer to human).

Interpretation. The universal 0.333 peak confirms pseudogenes are, compositionally, "broken CDS" — consistent with their origin. The magnitude of that peak is much more plausibly explained by **pseudogene annotation depth/completeness** than by biology: the human T2T/GENCODE pseudogene set is the most thoroughly curated of the seven, so its pseudogene SCFR pool is enriched for well-defined, relatively young, still highly triplet-structured pseudogenes, while shallower non-human annotations likely mix in older, more degraded, or mis-annotated calls that dilute the signal.

Speculation (flagged). It is tempting to read "sharper pseudogene peak = younger pseudogene population" and build an age argument from this, but that would conflate annotation-quality differences with biological age differences. This cannot be resolved from the pooled density plot alone — see Section 7 for the check (pseudogene age/divergence stratification per species).

2.3 Purely intergenic SCFRs

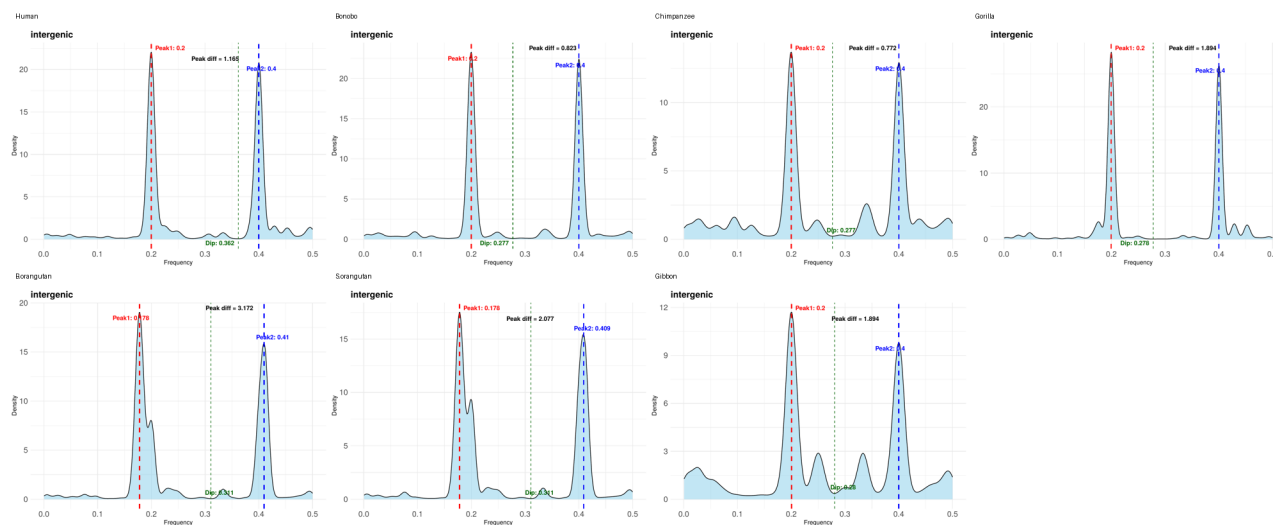


Figure 3. Intergenic SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.200	0.400	0.362	1.17
Bonobo	0.200	0.400	0.277	0.82
Chimpanzee	0.200	0.400	0.277	0.77
Gorilla	0.200	0.400	0.278	1.89
B. orangutan	0.178	0.410	0.311	3.17
S. orangutan	0.178	0.409	0.311	2.08
Gibbon	0.200	0.400	0.280	1.89

Observation. A completely different, and completely shared, signature from CDS: a sharp **twin peak at 0.20 and 0.40** in every species (both peaks are comparably tall — "Peak diff" here is small, meaning neither peak dominates the other), with essentially **no density at 0.333**. Both orangutan species show a small but consistent leftward shift of the low-frequency peak, to 0.178 instead of 0.200.

Interpretation. As noted in Section 1, a 0.2/0.4 pair is the DFT fingerprint of an underlying **5-bp periodic repeat**, not codon structure. This is the expected outcome for bulk intergenic DNA, which genome-wide is dominated by simple tandem repeats, satellite DNA, and transposable-element-derived repeats with strong internal short-period structure — not by protein-coding constraint. This is a **shared, deeply conserved primate-wide pattern**: intergenic SCFRs are, in bulk, "repeat-periodic" rather than "codon-periodic," and this holds across the whole hominoid clade.

Interpretation of the orangutan shift. A period shift from 5.0 bp ($f=0.200$) to ≈ 5.6 bp ($f=0.178$) is small but reproducible in both orangutan species and not seen in any African ape or gibbon, suggesting a genuinely orangutan-lineage-specific repeat-unit-length difference in the intergenic repeat elements that dominate this pool (e.g. a lineage-specific satellite/microsatellite variant), rather than noise — the same shift appearing independently in two related genomes argues against a one-off artifact.

2.4 Intergenic SCFRs antisense to a genic strand

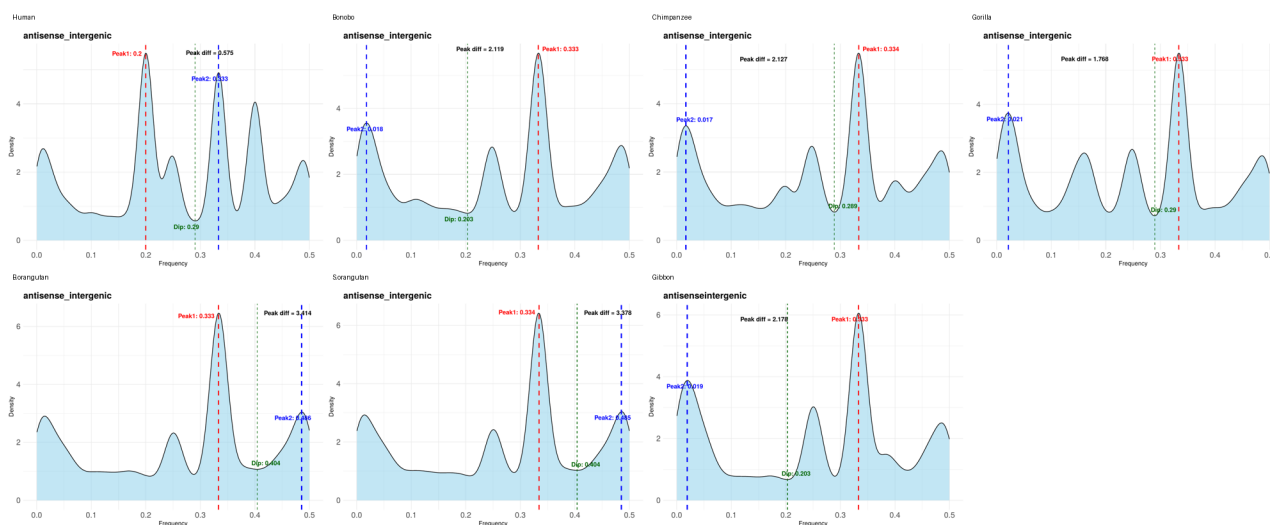


Figure 4. Antisense-intergenic SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.200	0.330	0.290	0.58
Bonobo	0.333	0.018	0.203	2.12
Chimpanzee	0.334	0.017	0.289	2.13
Gorilla	0.333	0.021	0.290	1.77
B. orangutan	0.333	0.486	0.404	3.41
S. orangutan	0.334	0.485	0.404	3.38
Gibbon	0.333	0.019	0.203	2.18

Observation. This is the broadest, flattest, most multi-modal category in the dataset — peak heights are only ≈ 3 – 7 (versus ≈ 20 for CDS and intergenic), and every panel shows 3–4 separate bumps rather than one or two clean peaks. **Six of seven species (all but human) have their single tallest peak at 0.333–0.334** — the codon position — while human's tallest peak sits at 0.20 instead, more like bulk intergenic DNA. A small, consistent near-zero bump (~ 0.02) is present in every non-orangutan species.

Interpretation. "Antisense to a genic strand" sequence is, by definition, physically embedded near or overlapping a transcribed locus, on the opposite strand. That proximity to active transcription plausibly explains why this pool, unlike bulk intergenic DNA, shows the codon periodicity as its dominant (not merely present) signal in most species — even though these sequences are not themselves annotated as coding. The much lower absolute peak heights than CDS confirm the signal is real but far weaker/less universal than true coding sequence: only a sub-population of

this pool carries strong 0.333 periodicity, while the rest contributes the broad, low background and the low- and high-frequency side bumps.

Speculation (flagged), and why it matters for the proto-gene hypothesis. The human-specific shift away from 0.333 and toward the 0.20 intergenic-like signature is the single most striking cross-species anomaly in this dataset. Two very different explanations are both consistent with what's visible here, and only raw data can distinguish them: (i) a genuine biological difference — human antisense-intergenic sequence may be more repeat-dominated (Alu/LINE insertion density near genes differs between human and other apes) than the equivalent regions in other apes; or (ii) an annotation artifact — if the "genic strand" annotation used to define antisense is more complete/different in human (again due to annotation depth), the resulting SCFR pool could differ in composition for reasons unrelated to sequence biology. This needs the raw-data check in Section 7 before treating it as a real signal.

2.5 Intron-overlapping SCFRs

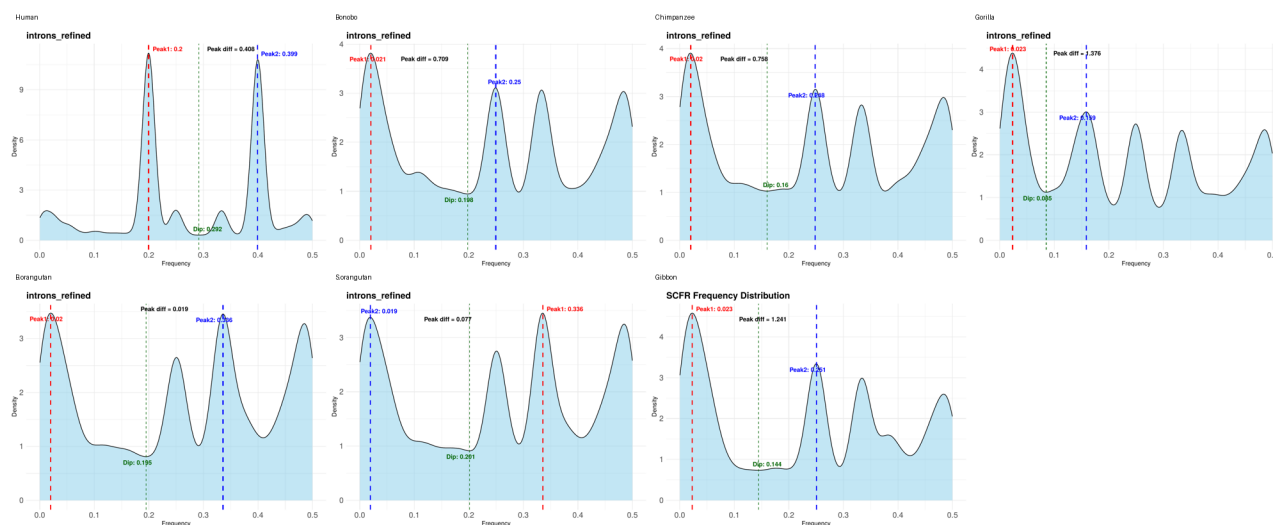


Figure 5. Intron-overlapping SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.200	0.399	0.292	0.41
Bonobo	0.021	0.250	0.198	0.71
Chimpanzee	0.020	0.208	0.160	0.76
Gorilla	0.023	0.159	0.085	1.38
B. orangutan	0.020	0.336	0.195	0.02
S. orangutan	0.336	0.019	0.201	0.08
Gibbon	0.023	0.251	0.144	1.24

Observation. The *broadest and flattest* density curves of any category (peak heights only ≈ 1 –5, roughly a quarter of CDS/intergenic) — consistent with introns being the single least compositionally constrained category tested. All six non-human apes share a near-zero dominant peak (0.02–0.023, a slowly-varying compositional trend, not a short repeat), but their *secondary* peak location is scattered across a wide range: 0.159 (gorilla), 0.208 (chimpanzee), 0.250–0.251 (bonobo, gibbon), and 0.336 (both orangutans, where the near-zero/codon-adjacent peaks are almost exactly tied — Peak diff ≈ 0.02 –0.08, the smallest in the dataset). Human again breaks from every other species, showing the intergenic-like 0.2/0.4 twin peak instead.

Interpretation. The near-zero dominant peak reflects the well-established fact that introns are the least sequence-constrained of the transcribed categories, with broad, slowly-varying isochore-like GC/AT composition dominating over any short repeat or codon signal. The wide scatter in *where* the weaker secondary periodicity falls

(0.159 to 0.336) across otherwise closely related species is itself informative: it suggests intron composition retains substantial lineage-specific repeat content (different transposable-element insertion histories per species) layered on top of a shared, weak background trend, rather than one shared secondary process common to all apes.

Speculation (flagged). The orangutan-specific near-tie between the near-zero peak and the 0.336 (codon-adjacent) peak is worth flagging as a candidate signature of a larger-than-usual codon-periodic sub-population within orangutan introns — plausibly retained/alternatively spliced intronic sequence exposing a nascent ORF to weak selection. This is exactly the kind of "partial convergence toward 0.333" your proto-gene hypothesis predicts, but it could equally be a smaller total intron SCFR pool for orangutans making the secondary bump look proportionally larger by chance — needs confirmation (Section 7).

2.6 UTR-overlapping SCFRs

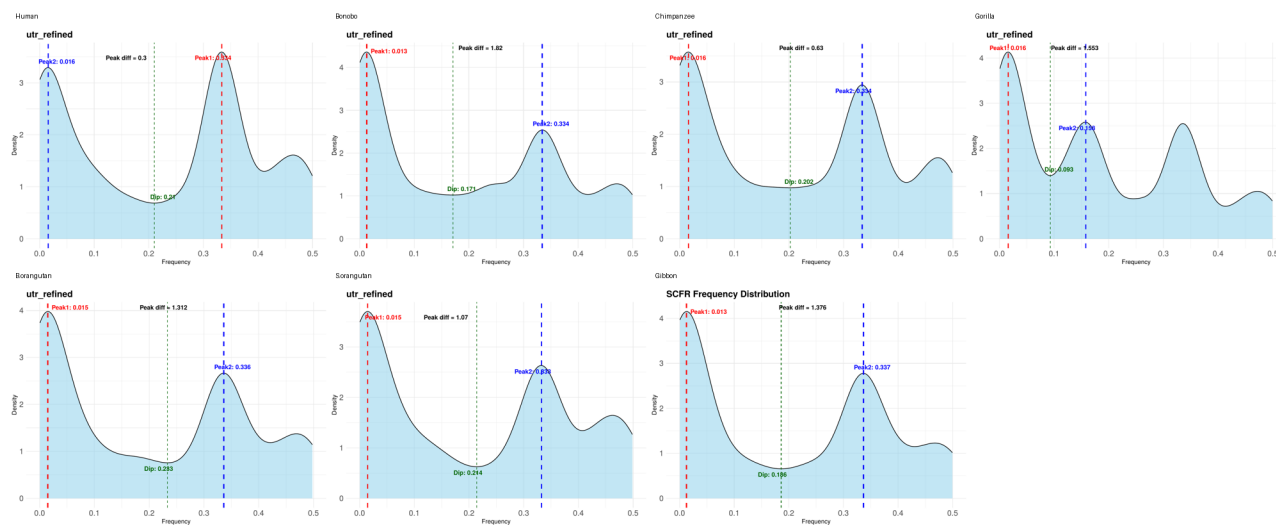


Figure 6. UTR-overlapping SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.334	0.016	0.210	0.30
Bonobo	0.013	0.334	0.171	1.82
Chimpanzee	0.016	0.334	~0.20	0.63
Gorilla	0.016	0.158	0.093	1.55
B. orangutan	0.015	0.336	0.283	1.31
S. orangutan	0.015	~0.33	0.214	1.07
Gibbon	0.013	0.337	0.186	1.38

Observation. A consistent, low-amplitude bimodal shape (peak heights only ≈ 2.5 – 4.5) with a near-zero compositional-trend peak and a much weaker codon-adjacent bump. Human is the only species where the two peaks are close to equal height, with the 0.334 side edging out the near-zero side; in every other species the near-zero peak is clearly taller. Gorilla again shows an intermediate secondary peak (0.158) rather than the codon position.

Interpretation. UTRs are transcribed but (mostly) not translated, so the presence of any 0.33 signal at all is attributable to a minority of upstream/downstream ORF-adjacent sequence or short upstream ORFs (uORFs) known to occur in many UTRs, sitting on top of a majority background of non-coding, compositionally drifting sequence (the near-zero peak). Human's comparatively stronger codon-adjacent signal is consistent with human UTR annotation capturing proportionally more uORF-containing or CDS-adjacent UTR sequence, but, as with pseudogenes, an annotation-depth explanation cannot be ruled out without per-sequence data.

2.7 RNA-gene-overlapping SCFRs

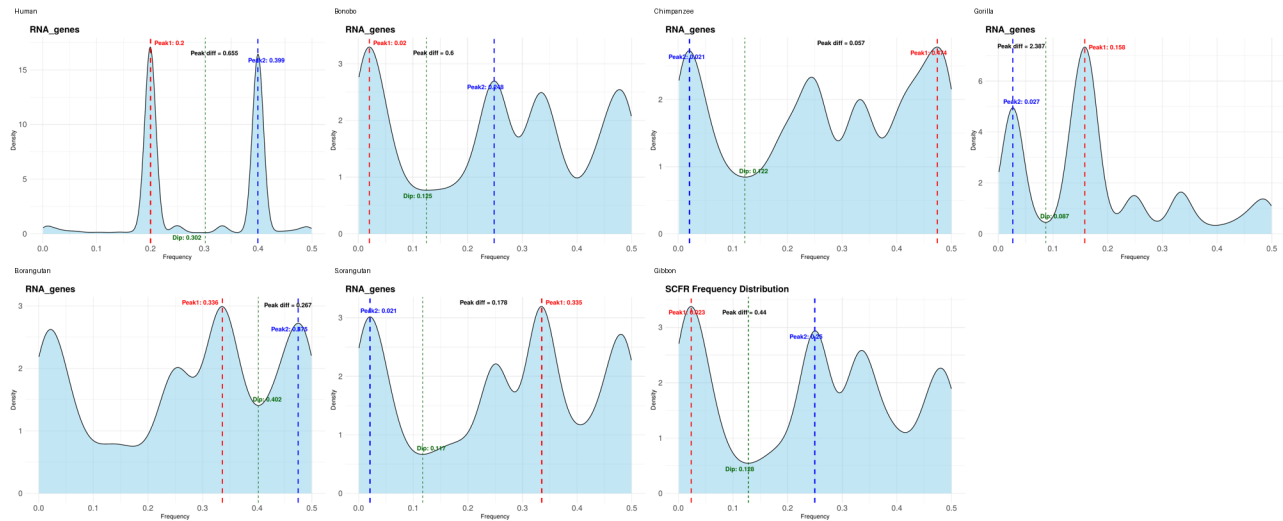


Figure 7. RNA-gene-overlapping SCFR frequency density, all 7 species.

Species	Peak1 (freq)	Peak2 (freq)	Dip	Peak diff
Human	0.200	0.399	0.302	0.66
Bonobo	0.020	0.248	0.125	0.60
Chimpanzee	0.474	0.021	0.122	0.06
Gorilla	0.158	0.027	0.087	2.39
B. orangutan	0.336	~0.17	0.402	0.27
S. orangutan	0.335	0.021	0.117	0.18
Gibbon	0.023	~0.25	0.128	0.44

Observation. This is the **least consistent category in the entire dataset** — every single species shows a different dominant frequency (0.02, 0.16, 0.20, 0.336, 0.474 are all seen as someone's Peak1), and peak dominance is uniformly weak (Peak diff mostly <1, versus >18 for CDS) — i.e., no species shows one periodicity clearly winning out over the others. Chimpanzee in particular shows a near-Nyquist peak at 0.474 (period ≈ 2.1 bp) essentially tied in height with three other bumps across the spectrum.

Interpretation. "RNA genes" is the most compositionally heterogeneous category by definition — it lumps together rRNA, tRNA, snRNA, snoRNA, miRNA host sequence, and long non-coding RNA (lncRNA) loci, which have very different base-composition and repeat profiles from one another. It is also very likely the smallest SCFR pool of the seven categories in most species (consistent with the human count of 234,829 out of 3.23M total ≥ 300 bp SCFRs, versus 1.5M for intergenic), so the pooled density estimate here is expected to be noisier and more sensitive to whichever few sub-types of RNA gene happen to be well-represented in that species' annotation. Both effects together plausibly explain the lack of any shared cross-species pattern, without needing a deeper biological explanation.

Why this category still matters for proto-genes (flagged as speculation). Long non-coding RNAs are exactly the substrate the "already transcribed, not yet translated" stage of your model describes, and a growing literature documents lncRNA loci that harbor short, translated open reading frames (micropeptides) under weak-to-moderate selection. The orangutan and B. orangutan 0.336 dominant peaks in this category are consistent with (but not proof of) a sub-population of RNA-gene-overlapping SCFRs acquiring codon periodicity — i.e., candidate proto-genes hiding inside currently-annotated non-coding RNA loci. Confirming this requires isolating that sub-population from the rest of the (very heterogeneous, very noisy) pool, not reading it off the pooled curve. See Section 7.

3. Cross-category and cross-species synthesis

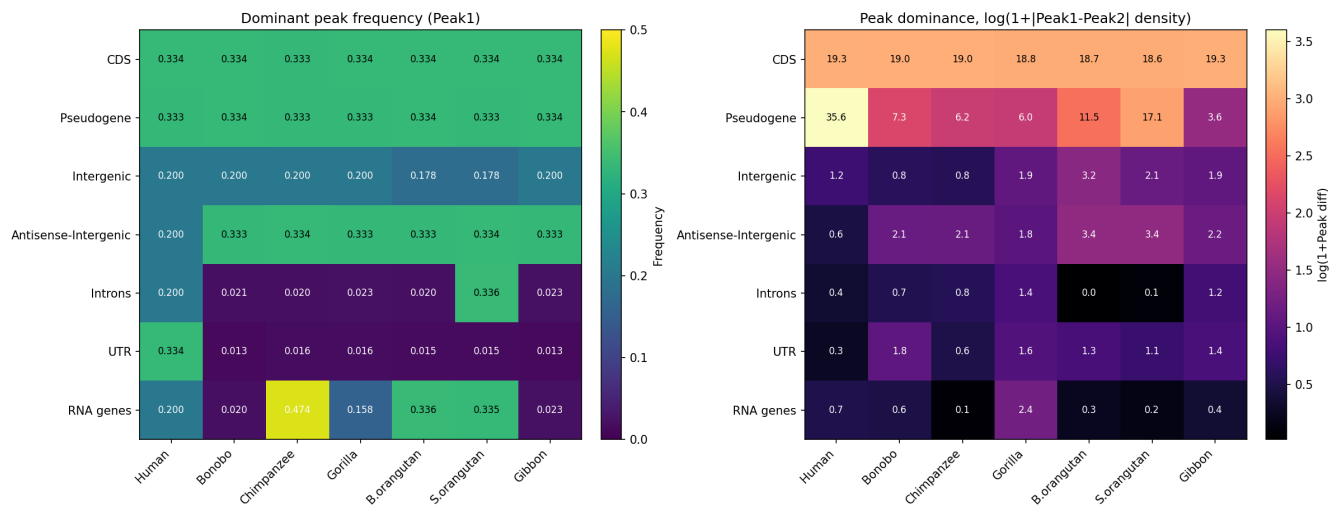


Figure 8. Left: dominant (Peak1) frequency for every category×species combination. Right: peak dominance (how much taller Peak1 is than Peak2, log-scaled) — a proxy for how strongly one periodicity dominates the pooled population versus a heterogeneous mixture.

3.1 A constraint gradient, not a binary split

Ranking categories by peak dominance (Fig. 8, right panel) gives a clear, ordered gradient rather than a sharp coding/non-coding divide:

- **CDS (Peak diff ≈ 18.6 – 19.3)** — overwhelming, near-total dominance of one periodicity. The reference point for "fully coding-constrained."
- **Pseudogene (Peak diff ≈ 3.6 – 35.6 , species-dependent)** — strong dominance, but consistently weaker than true CDS and far more variable across species.
- **Intergenic and antisense-intergenic (Peak diff ≈ 0.6 – 3.4)** — moderate, roughly comparable dominance, but of *different* periodicities (repeat-driven 0.2/0.4 for bulk intergenic; codon-adjacent 0.333 for most species' antisense-intergenic).
- **UTR, introns, RNA genes (Peak diff mostly < 1.5 , often < 0.5)** — the weakest, flattest, most heterogeneous distributions: no single periodicity meaningfully dominates the pool.

This gradient is exactly the shape predicted by a **continuum model of sequence constraint**: categories under strong, uniform, long-standing selection (CDS) converge tightly on one signal; categories under no coding constraint at all but strong repeat content (intergenic) converge tightly on a *different* signal; and categories that are transcribed-but-mostly-unconstrained, or definitionally heterogeneous (introns, UTRs, RNA genes), sit in between with weak, mixed, unstable signals. It does *not*, by itself, distinguish your specific "proto-gene spectrum" hypothesis from a simpler explanation (these categories are just genuinely heterogeneous mixtures of many unrelated sequence types with no shared biology) — see Section 5 for why, and Section 7 for the test that can tell the two apart.

3.2 Shared patterns across species

- The CDS 0.333–0.334 peak location and its near-total dominance is essentially invariant across all 7 species — the only fully universal, non-negotiable signal in the dataset.
- The intergenic 0.2/0.4 twin-peak (repeat-periodicity) signature is shared by all 7 species, with a small but consistent orangutan-specific frequency shift (0.178 vs 0.200).
- The pseudogene 0.333 peak location (though not its height) is shared by all 7 species.

- A near-zero (compositional-trend) peak recurs as the dominant or co-dominant feature in introns and UTRs across almost every species — the shared signature of "transcribed but not under coding constraint."

3.3 Unique / lineage-specific patterns

- **Human breaks the ape-wide pattern in three separate categories** (introns, antisense-intergenic, and RNA genes), each time showing the intergenic-like 0.2/0.4 signature where every other ape shows a different dominant peak (near-zero or 0.333). This is the most consistent single anomaly in the whole dataset and recurs too often across independent categories to dismiss as one-off noise — but it is exactly the kind of pattern that a difference in annotation pipeline/version (human genomes are typically the most deeply and differently annotated of the seven) can also produce, so it needs confirmation before being read as biology (Section 7).
- **Both orangutan species consistently diverge together** from the African apes and gibbon: the shifted 0.178/0.41 intergenic peak, and the near-tied near-zero/0.336 intron peaks not seen in any other species. Because the two orangutan species (Bornean and Sumatran, a relatively recent split) agree with each other and disagree with everyone else, this looks like a genuine orangutan-lineage signal rather than noise.
- **Gorilla shows an idiosyncratic intermediate secondary peak** (0.158–0.159) in three different categories (CDS's mislabeled Peak2, UTR, RNA genes) where other species show either the codon peak or the near-zero peak — worth checking whether this reflects something specific to the gorilla T2T assembly or annotation rather than three independent biological coincidences.
- RNA genes show no shared cross-species pattern at all (Section 2.7) — the one category where "unique to each species" is the pattern, most plausibly because it is the smallest and most internally heterogeneous pool.

3.4 Breadth, not just peak location

Peak *height* (equivalently, breadth/spread, since each curve integrates to 1) carries information the peak location alone does not: it tells you what fraction of the pooled SCFR population actually shares the dominant periodicity, versus how much of the pool is compositionally unrelated "background." Two categories can have their peak at the same frequency and mean very different things:

- CDS and pseudogene both peak at 0.333, but CDS's peak is far taller and narrower — essentially the entire CDS pool is coding-like, while only a fraction of the pseudogene pool retains that signal as strongly, with the rest smeared across the broad, low background visible under the pseudogene peaks in Fig. 2.
- Antisense-intergenic and RNA genes both sometimes peak near 0.333, but their peak heights (≈ 3 –7 and often < 3 respectively) are 3–7× lower than CDS — the periodicity is present as the *most common single value*, but it is far from universal within the pool, and a large fraction of each pool sits elsewhere in the spectrum.
- Introns and UTRs have the lowest peak heights of all — the broadest, flattest distributions — meaning these pools are the most compositionally diverse of the seven categories, with no periodicity, codon-related or otherwise, describing more than a small minority of the sequences.

Practically: **breadth is a direct proxy for how mixed a category is**, and the categories with the broadest curves (introns, UTR, RNA genes, and to a lesser extent antisense-intergenic) are exactly the categories where a real proto-gene sub-population would be easiest to hide inside an aggregate plot, and hardest to see without going back to per-sequence data.

4. A structural caveat: pool size and the dilution effect

The human ≥ 300 bp SCFR counts reported in your background note are a useful sanity check on all of the above:

Category	SCFR count (human, ≥ 300 bp)	Share of total
Intergenic	1,525,405	47.2%
Antisense-intergenic	931,113	28.8%
Introns	1,372,258	42.5%
RNA genes	234,829	7.3%
UTR	47,593	1.5%
Pseudogene	21,994	0.7%
CDS	29,733	0.9%
Total (all categories)	3,232,691	100%

(Percentages are of the 3,232,691 total; categories overlap so they need not sum to 100%.)

The categories with the sharpest, most homogeneous signal (CDS, pseudogene) are also, by a wide margin, the *smallest* pools; the categories with the broadest, most heterogeneous pooled signal (intergenic, introns, antisense-intergenic) are the *largest*, by 1–2 orders of magnitude. This matters directly for the proto-gene question: if even 0.1–1% of the ~1.4 million intronic or ~1.5 million intergenic SCFRs were genuine emerging proto-genes with a real 0.333 signal, that sub-population (a few hundred to a few thousand sequences) would be almost completely invisible in a pooled density plot dominated by over a million unrelated sequences — it would show up, at most, as a barely perceptible shoulder, not a peak. **The pooled plots in this dataset are therefore not well suited to detecting a rare proto-gene sub-population directly**, and their flat, broad shape in the large non-coding categories should not be read as evidence *against* the existence of such a sub-population — only as evidence that it isn't the majority of the pool, which was never in question.

5. What could represent proto-genes in this data

Given the staged model in your background note — (1) acquire transcription, (2) acquire a long enough ORF to be translated, exposing the sequence to coding-level selection, (3) progressively accumulate stronger triplet/frame structure — the categories that are *already transcribed* are the only plausible source of true proto-genes; a sequence with no transcription cannot yet be exposed to the translational selection that would generate a 0.333 signal in the first place. That immediately reorders the candidates:

5.1 Best candidates: transcribed, non-coding categories with a partial 0.333 signal

- **Antisense-intergenic SCFRs** (Section 2.4). Six of seven species already show the 0.333 peak as their single tallest peak — weaker and broader than CDS, but real and dominant. These sequences sit antisense to a genic locus, meaning local RNA polymerase activity (antisense transcription, transcriptional read-through, or bidirectional promoters) is a plausible, biologically well-documented route to step (1) of the model without requiring a whole new promoter to evolve from scratch.
- **RNA-gene-overlapping SCFRs** (Section 2.7), specifically the orangutan sub-population showing a dominant 0.335–0.336 peak. RNA genes are, by definition, already transcribed (that's the annotation), so any sequence within this category that also shows a codon-adjacent periodicity has already cleared step (1) and is showing early evidence of step (2–3). This maps directly onto the documented phenomenon of small/micropeptide-encoding open reading frames emerging within long non-coding RNA loci.
- **Intronic SCFRs with a secondary codon-adjacent bump** (Section 2.5), particularly the orangutan 0.336 co-dominant peak. Intronic sequence is transcribed as part of the pre-mRNA by definition; retained-intron transcripts or alternatively-spliced isoforms are a documented route by which an intronic ORF becomes translated occasionally, providing a weak but real selective pressure.

5.2 Weaker or unlikely candidates

- **Bulk (non-antisense) intergenic SCFRs** are the least likely direct source: their dominant signal (0.2/0.4) reflects short tandem-repeat/satellite structure, which is compositionally very far from codon structure and is not obviously a transitional state toward it — and, more basically, most of this pool has no established route to transcription at all, which is the first requirement of the model. Any proto-genes hiding here are most plausibly a small minority that has separately gained a promoter/transcription unit and would look nothing like the repeat-dominated bulk signal — exactly the kind of rare sub-population the dilution effect in Section 4 would hide.
- **UTRs** already have an assigned function (transcript stability, localization, translational regulation) and are physically attached to a real gene; a codon-like signal here more likely reflects known upstream ORFs (uORFs) than emerging de novo proto-genes, though the two are not mutually exclusive.

5.3 Does the data support a spectrum, or a small number of near-CDS outliers?

The pooled density plots, as generated, **cannot distinguish these two possibilities**, and this is the central methodological limitation of the current analysis relative to your hypothesis. A true continuous spectrum of proto-gene maturity (many sequences at many intermediate stages between "no signal" and "CDS-like") and a mixture of two discrete populations (a large majority with no coding signal, plus a small minority already fully CDS-like, with nothing in between) can produce very similar-looking *pooled* density curves — both would show a broad low background plus a possible smaller bump near 0.333, and only the per-sequence distribution of individual peak amplitudes at exactly $f=0.333$ can tell them apart. This is the single highest-value next analysis step, detailed in Section 6.

6. How to confirm candidate proto-genes

In roughly increasing order of effort, and building on data you already have:

Step 1 — Stop pooling; look at the per-sequence distribution

For each candidate category (antisense-intergenic, RNA genes, introns), extract each individual SCFR's amplitude specifically at $f=0.333$ (not just its single dominant frequency) from the summary.tsv already produced for the R plotting scripts. Plot the distribution of these per-sequence amplitudes. A spike of high-amplitude sequences clearly separated from a bulk of near-zero-amplitude sequences supports a two-state (discrete) model; a smooth, continuously populated range of amplitudes between the two extremes supports your spectrum model. This directly answers the question posed in Section 5.3 and requires no new sequence analysis — only a different view of data you already have.

Step 2 — Pull out and characterize the high-amplitude tail

- Rank SCFRs within each candidate category by amplitude at $f=0.333$ and pull the top percentile (e.g., top 0.1–1%) as a candidate proto-gene set.
- Check ORF length and completeness within each candidate (start codon presence, length distribution) — a real proto-gene needs a sufficiently long, uninterrupted reading frame, which is a stronger requirement than "no in-frame stop for 300bp."
- Check GC content and dinucleotide composition against genome-wide background for the same species — periodicity-3 signals can arise from strong GC content skew alone, independent of any reading-frame selection, so this needs to be ruled out with an appropriate composition-matched null (e.g., shuffled or Markov-model-generated control sequences of matching length and GC content, then rerun through the same DFT pipeline).

Step 3 — Independent lines of evidence per candidate locus

- **Transcription evidence:** overlap candidate SCFR coordinates with RNA-seq / CAGE data (if available for these species/tissues) to directly confirm they are transcribed, rather than inferring it from category membership alone.
- **Translation evidence:** overlap with ribosome-profiling (Ribo-seq) data where available — this is the most direct possible confirmation that a candidate proto-gene is actually being translated, not just compositionally coding-like.
- **Selection evidence:** where cross-species alignments exist for a candidate locus, check dN/dS-style substitution patterns; genuine emerging coding constraint should show reduced nonsynonymous substitution relative to a matched non-coding control, even at an early stage.
- **Cross-species locus tracing:** for a candidate locus found in one species, check whether the orthologous region in an outgroup species shows weaker or absent periodicity, and whether more closely related species show intermediate signal. A consistent phylogenetic gradient at the same locus (rather than genome-wide aggregate) is much stronger support for the staged/spectrum model than any genome-wide pooled comparison can provide, and directly operationalizes the "different stages of de novo gene birth" hypothesis at the level of individual loci rather than whole categories.

Step 4 — Resolve the annotation-depth confound

Several cross-species differences flagged above (human's repeated divergence from other apes; the large pseudogene peak-height range; possible gorilla-specific quirks) are equally consistent with differing annotation completeness between species as with real biology. Before drawing conclusions from any single-species anomaly, check whether the relevant annotation (pseudogene calls, gene/genic-strand definitions used for the antisense category, RNA gene sets) was generated with the same pipeline, source database, and level of curation across all 7 species. Where annotation provenance differs, treat cross-species differences in that category as unconfirmed until

checked against a harmonized annotation set.

7. Limitations of this pass

- This report is based on visual inspection of rendered PDF plots and the peak/dip values already annotated on them by your R scripts — it is not a re-analysis of the underlying per-sequence frequency/amplitude data, which would allow exact statistics (means, confidence intervals, formal peak-width/FWHM measurements) rather than the approximate, plot-read values reported here.
- A handful of peak labels in the source PDFs were small enough that exact values could only be estimated (marked with "~" or "*" in the tables above); these are flagged rather than silently treated as exact.
- The auto-annotated "Peak1"/"Peak2" labels do not consistently mean "tallest"/"second-tallest" across every plot (e.g., gorilla's CDS Peak2 landed on a minor ripple, not the visible ~0.43 shoulder); where this was evident it is called out explicitly rather than treated as ground truth.
- "Peak diff" is inferred here to be the density-height gap between Peak1 and Peak2 (this is consistent with every value checked against the corresponding plot); it is not independently documented and should be confirmed against your R script's actual definition.
- No statistical testing (e.g., are cross-species peak-location differences significant, or within the range expected from finite-sample noise) has been performed; the observations above describe what is visible in the plots, not confirmed statistical differences.

End of report.